VIJAYABHERI

Malappuram District Panchayat Educational Project



LEAP-26

ENGLISH MEDIUM

MATHEMATICS

SSLC QUESTION POOL

VIJAYABHERI

MALAPPURAM DISTRICT PANCHAYAT
EDUCATIONAL PROJECT

LEAP 26

SSLC MATHEMATICS QUESTION POOL

ആമുഖം

2001–2002 വർഷത്തിൽ മലപ്പുറം ജില്ലയുടെ സമഗ്ര വിദ്യാഭ്യാസ പുരോഗതി ലക്ഷ്യം വെച്ച മലപ്പുറം ജില്ലാ പഞ്ചായത്ത് ആസൂത്രണം ചെയ്ത നടപ്പിലാക്കിവരുന്ന പദ്ധതിയാണ് വിജയഭേരി വിദ്യാഭ്യാസ പദ്ധതി.

കഴിഞ്ഞ 24 വർഷത്തിനുള്ളിൽ ജില്ലയുടെ വിദ്യാഭ്യാസ മുഖച്ഛായ മാറ്റാൻ ഈ പദ്ധതിയിലൂടെ കഴിഞ്ഞിട്ടുണ്ട്. എസ്എസ്എൽസി വിജയശതമാനത്തിലും പ്ലസ് ടു വിജയശതമാനത്തിലും മത്സര പരീക്ഷകളിലും ഏറെ പിന്നിലായിരുന്ന ജില്ല ഇന്ന് സംസ്ഥാനത്തെ മറ്റേത് ജില്ലയോടും കിടപിടിക്കാവുന്ന രീതിയിൽ മാത്രമല്ല, പല മേഖലയിലും അവരെക്കാൾ മുന്നേറാൻ ഇന്ന് ജില്ലയ്ക്ക് കഴിഞ്ഞിട്ടുണ്ട്. ഏറ്റവും കൂടുതൽ എ പ്ലസ്സുകളുള്ള ജില്ല, ഏറ്റവും കൂടുതൽ കുട്ടികൾ കേന്ദ്ര സർവകലാശാലയിലേക്ക് പ്രവേശനം നേടുന്ന ജില്ല, മെഡിക്കൽ എൻജിനീയറിങ് എൻട്രൻസ് പരീക്ഷകളിൽ മുനിൽ നിൽക്കുന്ന ജില്ല ഇന്ന് മലപ്പുറം ജില്ലയാണ്. വിദ്യാഭ്യാസപരമായി പിന്നാക്കം നിൽക്കുന്ന ജില്ല എന്ന അപഖ്യാതി നമ്മളിന് മറികടന്നു.

മറ്റ ജില്ലകളിൽ നിന്ന് ഭിന്നമായി ജില്ലയിലെ എല്ലാ തദ്ദേശ സ്വയംഭരണ സ്ഥാപനങ്ങൾക്കും എംഎൽഎമാർക്കും ഇന്ന് പ്രത്യേക വിദ്യാഭ്യാസ പദ്ധതികളുണ്ട്. ഇത്തരം എല്ലാ പദ്ധതികൾക്കും പ്രചോദനമായത് വിജയഭേരി വിദ്യാഭ്യാസ പദ്ധതിയാണ്. സംസ്ഥാനതലത്തിൽ മാത്രമല്ല ദേശീയ തലത്തിൽ പോലും ഈ പദ്ധതി ചർച്ച ചെയ്യപ്പെട്ട വരുന്നു. യൂനിവേഴ്സിറ്റികളിൽ വിജയിഭേരിപദ്ധതി ഗവേഷണ വിദ്യാർത്ഥികൾക്ക് പഠന വിഷയമാണ്.

ഈ പദ്ധതിയുടെ വിജയം മലപ്പുറം ജില്ലാ പഞ്ചായത്തിന്റെ വിജയമല്ല, ഇത് ഏറ്റെടുത്ത അധ്യാപകരുടെയും രക്ഷിതാക്കളുടെയും വിദ്യാർത്ഥികളുടെയും പിന്തുണയാണ് ഈ പദ്ധതിയുടെ വിജയത്തിന്റെ അടിസ്ഥാനം. അതുകൊണ്ടുതന്നെ അവരോടുള്ള കടപ്പാട് ഈ അവസരത്തിൽ പ്രത്യേകം രേഖപ്പെടുത്തുന്നു.

കഴിഞ്ഞ വർഷങ്ങളിലേതുപോലെ ഈ വർഷം നിലവിൽ വന്ന പുതിയ ജില്ലാ പഞ്ചായത്ത് ബോർഡും വിജയഭേരി പദ്ധതി കൂടുതൽ ഊർജ്ജസ്വലമായി മുന്നോട്ടുകൊണ്ടുപോകാൻ തീരുമാനിച്ചിട്ടുണ്ട്. അതിന്റെ ആദ്യപടിയായി ഈ വർഷം എസ്എസ്എൽസി പരീക്ഷ എഴുതുന്ന വിദ്യാർത്ഥികൾക്കും അധ്യാപകർക്കും സഹായകരമായ രീതിയിൽ എല്ലാ വിഷയങ്ങളുടെയും ചോദ്യശേഖരം പുറത്തിറക്കുകയാണ്.

പുതിയ പാഠപുസ്തകങ്ങളും പുതിയ രീതിയിലുള്ള പരീക്ഷ രീതിയുമാണ് ഈ വർഷത്തെ വിദ്യാർഥികൾ അഭിമുഖീകരിക്കേണ്ടത്. അവർക്ക് ഏറെ സഹായകമായ രീതിയിൽ റിവിഷൻ നടത്തുന്നതിനും പുതിയ രീതിയിലുള്ള ചോദ്യങ്ങൾ പരിചയപ്പെടുത്തുന്നതിനുമാണ് ഇത്തരമൊരു ചോദ്യശേഖരം തയ്യാറാക്കിയിരിക്കുന്നത്. എല്ലാ വിഷയങ്ങളുടെയും എല്ലാ പാഠങ്ങളിലുമുള്ള പ്രതീക്ഷിത ചോദ്യങ്ങൾ ഇതിൽ ലഭ്യമാക്കിയിട്ടുണ്ട്. ഈ ചോദ്യ ശേഖരം വിദ്യാർത്ഥികൾക്കും അധ്യാപകർക്കും ഏറെ സഹായകരമായിരിക്കുമെന്ന് പ്രതീക്ഷിക്കുന്നു.

ദൈനംദിന പ്രവർത്തനങ്ങൾക്കിടയിൽ ഇത്തരമൊരു ഭാരിച്ച ജോലി ഏറ്റെടുത്ത് കൃത്യസമയത്ത് പൂർത്തീകരിച്ച് പ്രസിദ്ധീകരിക്കുന്നതിന് ഞങ്ങളോടൊപ്പം നിന്ന അധ്യാപകരെ പ്രത്യേകം അഭിനന്ദിക്കുന്നു.

വൈസ് പ്രസിഡൻറ്
മലപ്പുറം ജില്ലാ പഞ്ചായത്ത്

പ്രസിഡന്റ്
മലപ്പുറം ജില്ലാ പഞ്ചായത്ത്

പി കെ അസ്സു
ആരോഗ്യ-വിദ്യാഭ്യാസ സ്റ്റാൻഡിംഗ് കമ്മിറ്റി ചെയർമാൻ
മലപ്പുറം ജില്ലാ പഞ്ചായത്ത്

ടി സലിം
കോഡിനേറ്റർ
വിജയഭേരി വിദ്യാഭ്യാസ പദ്ധതി

VIJAYABHERI MALAPPURAM DISTRICT PANCHAYATH

ശിൽപശാലയിൽ പങ്കെടുത്തവർ

UBAIDULLA KC
SOHSS AREEKODE

FIROZEKHAN K
GHSS OTHUKKUNGAL

SALEEM FAISAL P
GHSS OTHUKKUNGAL

THASNI BP
SOHSS AREEKODE

SAFREENA MT
PPTMYHSS CHERUR

KL JOSHY
MHS MOONNIYUR

ASMABI
GMVHSS VENGARA

SHARAFUDHEEN AREEKKADAN
GHS KURUKA

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1. ARITHMETIC SEQUENCES

1. Which of the following is not a term of the arithmetic sequence: 4,9,14,..... ?
[50, 51, 54, 44]

2 . Which of the following is a term of the sequence, 1,4,7,….?
(11 , 21, 31, 41)

3. Which of the following can be difference between two terms of the sequence, 3,7,11,15,… ?
(15, 19 , 20 , 21)

4. 7th term of an arithmetic sequence is 30. What is the sum of 6th and 8th terms?
(30 ; 44 ; 45 ; 60)

5. Second term of an arithmetic sequence is 25. What is the sum of first three terms?
(45 ; 50 ; 60 ; 75)

6. Some statements on the sequence 5,9,13,17,… are given.
Select the correct statements.
i) reminder on dividing the terms by five are equal
ii) 101 is a term of the sequence
iii) the terms of the sequence are two more than the multiples of3
iv) difference between two terms of the sequence can be 60
a) i , ii are true b) ii ,iii are true c) ii ,iv are true d) i,iii are true

7.Read the two statements below:
Statement A: If the fifth term of an arithmetic sequence is 8, then the sum of the third and 7th term is 16
Statement B: The sum of the two numbers got by adding and subtracting the same number from a number , is twice this number
Now chose the correct answer from those given below:
a) Statement A is true and statement B is false.
b) Statement A is false and statement B is true
c) Both statement are true, B is a reason for A
d) Both statement are true but B is not a reason for A

8. Read the two statements below:
Statement A: In an arithmetic sequence if the difference between 7th and 3rd term is 20 , then difference between 15th and 23rd term is 40
Statement B: In an athematic sequence, the change in terms is proportional to the change in position
Now chose the correct answer from those given below:
a) Statement A is true and statement B is false.
b) Statement A is false and statement B is true
c) Both statement are true, B is a reason for A
d) Both statement are true but B is not a reason for A

9. Read the two statements below:
Statement A: In an arithmetic sequence if the sum of the 1st and 14th term is 50, then the sum of the 6th and 9th term is 50
Statement B: In an arithmetic sequence if the sum of two positions is equal to the sum of the other two positions , then the sum of the terms at each pair is the same.
Now chose the correct answer from those given below:
a) Statement A is true and statement B is false.
b) Statement A is false and statement B is true
c) Both statement are true, B is a reason for A

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d) Both statement are true but B is not a reason for A

10. Read the two statements about the sequences 5,9,13, 17,…

Statement A: 100 can be difference between two terms of this sequence,

Statement B: Difference of any two terms of an arithmetic sequence is multiple of its common difference

Now chose the correct answer from those given below:

a) Statement A is true and statement B is false.
b) Statement A is false and statement B is true
c) Both statement are true, B is a reason for A
d) Both statement are true but B is not a reason for A

11. The fifth term of an arithmetic sequence is 70 and the 9th term is 50
    i) In this sequence by how much does each term decrease, when the position increases by one?
    ii) At what position does the term 0 occur in this sequence?
12. In the arithmetic sequence 8,14,20,. …
    a) can difference between any two terms, 90
    b) what is 18th term?
    c) Which term of the sequence is 200?
13. 17th term of an arithmetic sequence is 130 and 21st term is 160.
    a) Find 13th term?
    b)Find the sum of first 25 terms of the sequence?
14. 5th term of an arithmetic sequence is 40 and 12th term is 80. What is 26th term of the sequence? What is the sum of first 30 terms of the sequence?
15. Sum of first 15 terms of an arithmetic sequence is 600 and sum of first 25 terms is 1850.
    a) What is the 8th term of the sequence?
    b) What is the sum of first 20 terms of the sequence?
16. Sum of terms from 5th to 7th is 90. The sum of terms of 13th to 17th is 350.
    a) What is the 6th term of the sequence?
    b) What is the sum of first 20 terms of the sequence?
17. Common difference of an arithmetic sequence is 8 and 10th term is 21.
    a) Find its 11th term.
    b) What is the sum of first 20 terms of the sequence?
18. Sum of first 7 terms of an arithmetic sequence is 126 and the sum of first 14 terms is 448 .
    a) Find the 4th term
    b) write first three terms of the sequence
19. a) write the sequence of numbers which give reminder 1or 4 on division by 6
    b) Is this arithmetic sequence ? Why ?
    c) What is the 11th term of the sequence?
20. 4th term of an arithmetic sequence is 14.
    a) What is the sum of third and fifth terms?
    b) If second term is 6, what is sixth term?
    c) What is the common difference of the sequence?
21. Sum of first 5 terms of an arithmetic sequence is 55. The sum of first 10 terms of the sequence is 185 .
    a) What is the 3rd term of the sequence?
    b) What is 8th term of the sequence?
    c) What is the sum of first 15 terms of the sequence.
22. 9th term of an arithmetic sequence is 96 and 13th term is 72.
    a) What is 17th term of the sequence?

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b) What is 25th term ?

c) How many terms of the sequence starting from the first must be added to get zero as the sum?

23. Sum of all terms from 6th to 9th of an arithmetic sequence is 160.
Sum of terms from 11th to 14th is 240.
a) What is 10th term of the sequence?
b) What is 8th term?
c) What is the sum of first 4 terms of the sequence?

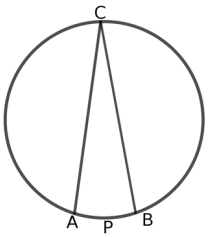
24. Sum of first 10 terms of an arithmetic sequence is 500. Fifth term of the sequence is 40
a) What is 6th term?
b) Write three different arithmetic sequence having sum of first 10 terms is 500

25. sum of first 12 terms of an automatic sequence is 468. And 19th term is 100 . Find the sum of first 25 terms of the sequence

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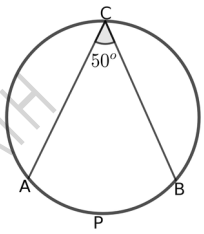
2.Circles and angles

1. In the picture A,B,P and Q are points on the circle. If arc APB is $\frac{1}{10}$ part of the circle. How much is <ACB?



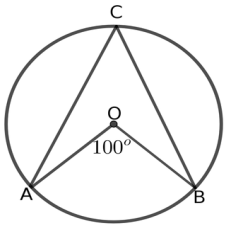
(10° ; 36° ; 72° ; 18°)

2. In the picture A,B,C and P are points on the circle. If <ACB = 50° . What fraction of the circle is the arc ABC.



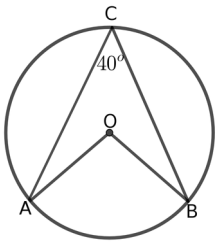
($\frac{5}{18}$; $\frac{5}{27}$; $\frac{5}{36}$; $\frac{5}{9}$)

3. In the picture O is the centre and A,B, C are points on the circle. *How much is <ACB ?*



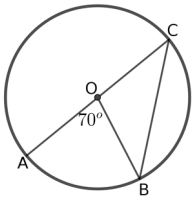
(100° ; 50° ; 75° ; 80°)

4. In the picture O is the centre and A,B, C are points on the circle. *How much is* <AOB ?



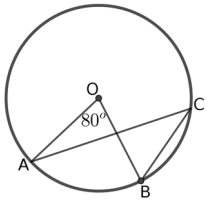
(140° ; 50° ; 80° ; 20°)

5. In the picture O is the centre and A,B, C are points on the circle. *How much is <ACB ?*



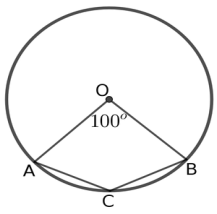
(140° ; 90° ; 35° ; 110°)

6. In the picture O is the centre and A,B, C are points on the circle. *How much is* <ACB ?



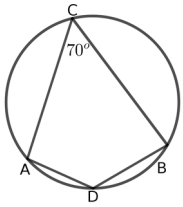
(100° ; 40° ; 90° ; 160°)

6. In the picture O is the centre and A,B, C are points on the circle. *How much is* <ACB ?



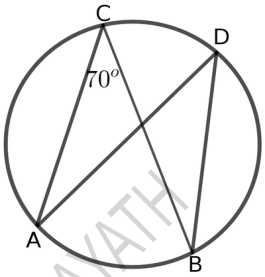
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(100° ; 130° ; 80° ; 260°)



7. In the picture A,B, C, and D are points on the circle. *How much is* < ADB ?

(35° ; 140° ; 110° ; 55°)



8. In the picture A,B, C and D are points on the circle. *How much is <ADB ?*

(35° ; 70° ; 140° ; 20°)

9. Read the statements below

Statement A: If all vertices of a quadrilateral are on a circle, then the sum of it's opposite angles is 180^0.

Statement B: In a circle the sum of the angles in alternate segments is 180^0.

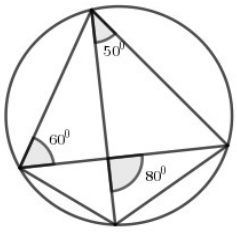
Choose the correct answer from those given below

a) A is true and B is Wrong

b) B is true and A is Wrong

c)Both are correct, B is the reason of A

d)Both are correct B is not the reason of A



10. Which of the following statements are related to the cyclic quadrilateral?

i)The sum of the opposite sides are equal

ii)The sum of the opposite angles are equal

iii)All sides are tangents to the same circle

iv)Four vertices are the points on the same circle

a) i and ii are correct b) iii and i are correct c) iv and iii are correct d) ii and iv are correct

11. Which of the among are cyclic quadrilateral ?

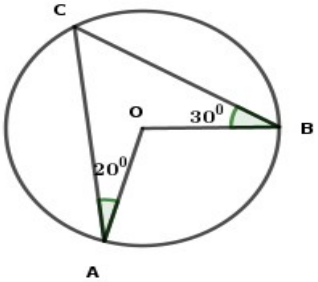
i) Parallelogram ii)Rectangle iii) Isosceles trapezium iv)Rhombus

a) i and ii are correct b) ii and iii are correct c) iv and iii are correct d) i and iv are correct

12. In the picture O is the center and A,B, C are points on circle. Which of the following are correct?

i) < ACB = 50^0 ii) < AOB = 50^0

iii) < ACB = 100^0 iv) < AOB = 100^0



a) i and ii are correct b) iii and ii are correct

c) iv and iii are correct d) i and iv are correct

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13. Read the statements below

Statement A : In cyclic quadrilateral, the outer angle at any vertex is equal to the inner angle at the opposite vertex.

Statement B : Sum of the linear pair is 180^0

Choose the correct answer from those given below

a) A is true and B is Wrong

b) B is true and A is Wrong

c)Both are correct, B is the reason of A

d)Both are correct B is not the reason of A

14. Read the statements below

Statement A : The angle obtained by joining the endpoints of a chord of central angle130^0 with another point on the circle is $55^{0.}$

Statement B: The angle made by joining the ends of an arc of a circle to any point on the alternate arc is half the central angle of the arc

Choose the correct answer from those given below

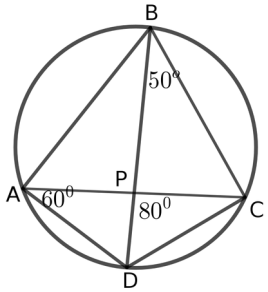
a) A is true and B is Wrong

b) B is true and A is Wrong

c)Both are correct, B is the reason of A

d)Both are correct B is not the reason of A.

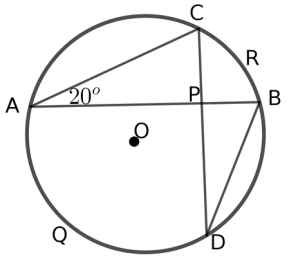
15. In the picture A, B, C, and D are points on circle and $< DAC = 60^0$, $< DBC = 50^0$, $< DPC = 80^0$ .Find the angles of the quadrilateral ABCD and the angles between the diagonals.



16 . In the picture O is the centre and A,B, C , D ,Q and R are points on the circle ;AB and CD are perpendicular to each other.

a) Find the angles of triangle PBC

b) Prove that the arcs AQD arcs BRC joined together make a semi-circle.



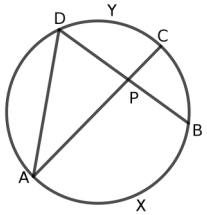
17. Draw a triangle with circumradius 3 centimetres and two of the angles 70° and 80°

18. Draw a triangle with circumradius 3.5 centimetres and two of the angles $52\frac{1^o}{2}$ and $62\frac{1^o}{2}$

19. Draw a triangle with circumradius 3 centimetres and two of the angles 40° and 110°

20. Draw a triangle with circumradius3 centimetres and two of the angles 50° and 40°

21. In the figure the central angle of arc AXB is 130° and the central angle of arc CYD is 70°. Find the angles of triangle APD



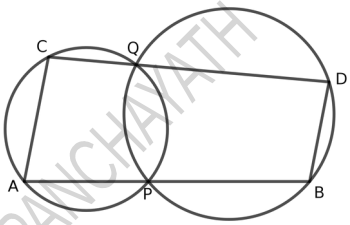
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22. Find the length of an arc of radius 10 cm and angles in alternate segment is 36° ?

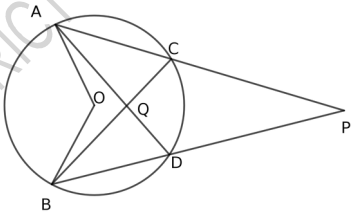
23. When the corner of a bent wire was placed at the centre of the circle $\frac{1}{6}$ of the circle was contained within it. If the corner of this wire is placed on a point of this circle, what fraction of the circle would it contains?

24. Draw a circle and a chord to split the circle into two parts in which all angles in one part must be 70° .

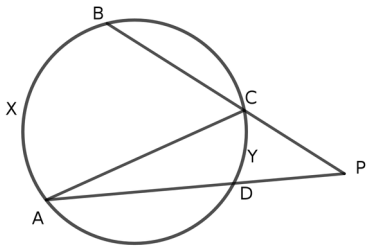
25. In the picture below, two circles intersects at P and Q and A,B, C are points on circle. Prove that if AB=CD then ABDC is cyclic quadrilateral.



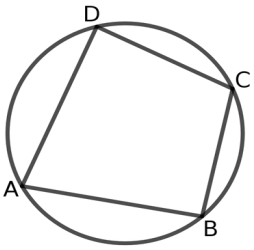
26. In the picture O is the centre and A,B, C , D ,Q and R are points on the circle. Prove that <AOB = <AQB + < APB



27. In the picture A,B, C and D are points on circle, and central angle of arc AXB and arc CYD are 110^0 and 30^0 respectively. Find the angles of triangle APC



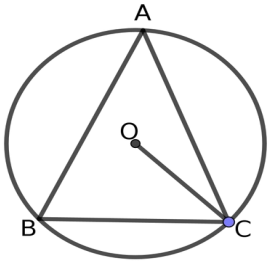
28. In the figure ABCD is cyclic quadrilateral. Prove that if AB = AD and CD = CB, then AC is diameter



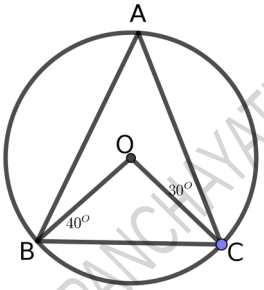
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29. In the picture O is the centre and A,B, and C are points on the circle. Prove that

< ABC + <OCA = 90°



30. In the picture O is the centre and A,B, and C are points on the circle and <OBC = 40° and < OCA = 30°. Find the angles of triangle ABC



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3. Arithmetic Sequence and Algebra

1. Algebraic form of an arithmetic sequence is 4n+3. What is the first term of the sequence?
(1, 3, 4 , 7)
2. Algebraic of form an arithmetic sequence is 6-4n. what is its common difference?
(6 , 2 , -4 , 4)
3. Algebraic of form an arithmetic sequence is 7n-4. What is the reminder obtained when the terms of the sequence are divided by 7 ?
(3, -3, 4, -4)
4. Sum of first 19 natural numbers is 190. What is the sum of first 20 natural numbers?
(200, 201, 210, 211)
5. Algebra of the sum of an arithmetic sequence is $7n^2$-2n. What is its first term?
(7 , -2 , 1 , 5)
6. Algebra of an arithmetic sequence is 8-3n. To get 49th term from 50th term
(3 is to be added, 3 to be subtracted, 8 is to be added, 8 is to be subtracted)
7. Algebra of an arithmetic sequence is 10n-8.
i) 1st term of the sequence is 10
ii) When the terms of the sequence are divided by we get the reminder 2
iii) All natural numbers, ending with 2 are the terms of the sequence
iv) All natural numbers, ending with 8 are the terms of the sequence
a) i , ii are true b) ii , iii are true c) ii , iv are true d) ii , iii are true
8. The sequence of the sums of the consecutive terms of an arithmetic sequence starting from the first is 3,12,27,48,..... what is the third term of the arithmetic sequence?
(9, 27, 15, 21)
9. First term of Arithmetic sequence is $\frac{4}{3}$ and the common difference is $\frac{1}{2}$. Show that there is no natural numbers as term in this sequence.
10. Algebra of a arithmetic sequence is 8n-4. Prove that the sum consecutive terms from the first will be always a perfect square
11. The sequence of the sums of the consecutive terms of an arithmetic sequence starting from the first is 4,15,33,58,….
a) Find the first term and the second term of the arithmetic sequence?
b) Write the algebra of the sum of terms of the sequence?
c) Write the algebra of the sequence?
12. Algebra of the sum of an arithmetic sequence is $3n^2$-4n.
a) Write the first term of the sequence?
b) Write the algebra of the sequence?
13. a) What is the sum of first 40 natural numbers?
b) What is the sum of first 40 terms of the sequence 6, 12, 18 , …...
c) What is the sum of first 40 terms of an arithmetic sequence with algebraic form 6n+4 ?
14. Algebra of arithmetic sequence is 2n+7.
a) What is the algebra of the sum of the sequence ?
b) Prove that when 16 is added to some of consecutive terms starting from the first gives a perfect square
15. What is the sum of the following arithmetic sequences?
a) 1 + 2 + 3 + 4 +……....+ 20
b) 4 + 8 + 12 + 16 +…… + 80

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c) 6 + 10 + 14 + 18 + ……..+ 82
d) 3 + 7 + 11 + 15 + ……..+ 79
e) 9 + 17 + 25 + 33 + ……+ 161

16. Algebra of arithmetic sequence is 5n+2.
    a) What is the common difference of the sequence?
    b) What is the 20th term of the sequence?
    c) What is the sum of first 20 terms of the sequence?
    d) How much more is the sum of first 20 terms of the arithmetic sequence with algebra 5n +3 than the sum of first 20 terms of the sequence 7,12, 17,22,…
17. a) Write the sequence of three digit numbers which give a reminder 2 when it is divided by 7. Is it an arithmetic sequence?
    b)Find the sum of all three digit numbers which leave a reminder 2 when it is divided by 7.
18. a) Sum of first 20 terms of an arithmetic sequence with a common difference, 6 is 1300. Write the sequence
    b) Write algebra of the sequence
    c) What is the algebra of the sum of the sequence?
19. Prove that the square of the terms of the arithmetic sequence 5,8,11,… is not a term of the sequence
20. Prove that the square of terms of the sequence 1,8,15,… are terms of the sequence
21. Prove that the square of any term of the arithmetic sequence 3, 9, 15, ... is the product of an odd number by 9.
22. Two arithmetic sequences are given below:
    5, 11, 17, ...
    8, 14, 20, ...
    a) What is the difference of the 4th terms of the two sequences?
    b) How much more is the sum of the first 25 terms of the second sequence, than the sum of the first 25 terms of the first sequence?
23. The first term of an arithmetic sequence is 4 and the common difference is 3.
    a) Calculate the sum of the first 10 terms of this sequence?
    b) Calculate the sum of the ten consecutive terms of this sequence from the second to the 11th?
    c) Can the sum of any 10 consecutive terms of the sequence be 500? Why?
24. Prove that the arithmetic sequence with first term $\frac{1}{3}$ and common difference $\frac{2}{3}$ contains all odd numbers. Prove also that this sequence contains no even numbers.
25. a) Calculate the sum of the first n terms of the arithmetic sequence with n^{th} term 5n+1
    b) How much more is the sum of the first 20 terms of the arithmetic sequence with n^{th} term 5n+4 than the sum of the first 20 terms of the sequence?
26. a) Write the algebraic form of the arithmetic sequence 7, 11, 15, ...
    b) Prove that no perfect square is a term of this sequence
27. Consider the arithmetic sequences given below.
    5, 8, 11, ...
    4, 7, 10, ...
    a) What is the reminder on dividing each term of the first sequence by 3?
    b) What is the reminder on dividing each term of the second sequence by 3?
    c) Prove that the square of each term of the first sequence is a term of the second sequence

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28. a) Is the sequence of natural numbers which leave an odd reminder on division by 6, an arithmetic sequence? What is the reason?
    b) Write the algebraic form of the sum of the first n terms of this sequence
29. 3, 6, 9, ...is a arithmetic sequence:
    a) What is the sum of first 20 terms?
    b) What is the difference of first 20 terms of the arithmetic sequence 3, 6, 9, ... and 6, 12, 18, ...
    c) What is the sum of first 20 terms of the arithmetic sequence 6, 12, 18, ...
30. Sum of first 5 terms of an arithmetic sequence is 65 and sum of first 20 terms is 860. Find its algebraic form
31. Consider the arithmetic sequence 1, 3, 5, ...and 1, 4, 7, ...
    a) Write the sequence of common terms of these sequences
    b) Is this sequence is arithmetic sequence? Why ?
    c) Find the 10th term of each three sequence
32. n^{th} term of an arithmetic sequence is 4n+1
    a) Find its 6th term
    b) Prove that this sequence contains many squares as terms
c) Prove that this sequence contains squares of all terms of this sequence

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4. MATHEMATICS OF CHANCE

1. a dice is rolled, then

   i) Probability of getting a prime number is $\frac{1}{2}$

   ii) Probability of getting a perfect square is $\frac{1}{2}$

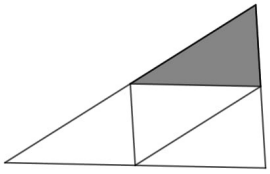
   iii) Probability of getting an odd number is $\frac{1}{2}$

   iv) Probability of getting a multiple of 5 is $\frac{1}{2}$

   a. (i)and(ii) are true b. (i)and(iii) are true

   c. (i)and(iv) are true d. (ii)and(iv) are true

2. The midpoints of the sides of a triangle is joined to get 4 small triangles. If we put a do on it without looking into it what is the probability of the dot falls on the shaded region?



A) $\frac{1}{2}$ B) $\frac{1}{4}$ C) $\frac{3}{4}$ D) $\frac{1}{3}$

3. A student is asked to say the first five prime numbers. What is the probability of saying an even number is ?

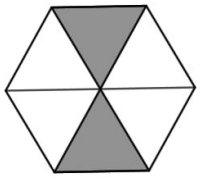
   (A) 0 (B) $\frac{2}{5}$ (C) $\frac{1}{5}$ (D) $\frac{4}{5}$

4. A person is asked to say a two digit number.

   What is the probability of getting two digits equal

   (A) 9% (B) 90% (C) 2% (D) 10 %

5. A regular hexagon is given in the picture



   Statement 1: If we put a dot in the figure the probability that the dot falls in the shaded region is 1/3

   Statement 2:all the 6 triangles are equilateral triangles with equal areas

   (A) statement 1 is true 2 is false

   (B) statement 1 is false 2 is true

   (C) statement 1& 2 is true statement 2 is the reason for statement 1

   (D) statement 1& 2 is true, statement 2 is not the reason for statement 2

6. In a box there are slips numbered 1,2,3,4 and in another box there are slips numbered -1,-2 and -3.if we take a number from each box, what is the probability of getting

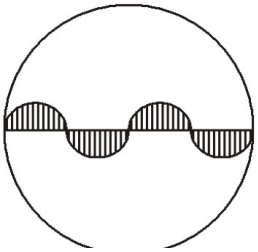
   (i) sum of the numbers as 0

   (ii) sum of the numbers is a Positive number

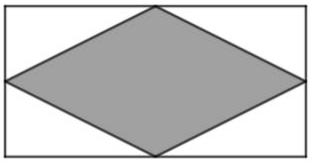
   (iii) sum of the numbers is a negative number

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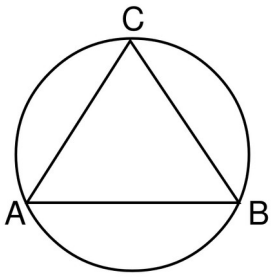
7. There are 7 red balls and 5 blue balls in a box. In another box there are 9 red balls and 7 blue balls
    a) If a ball is taken from the first box, what is the probability that it will gets red?
    b) What is the probability that it will red from the second box?
    c) If the all balls are put in a single box, what is the probability of getting a red ball?
8. The natural numbers from 10 to 25 were written on each piece of paper and put in a box. take a piece of paper from this.
    a) What is the probability that the number on the paper is an even number?
    b) What is the probability of becoming an odd number?
    c) What is the probability of becoming a Palindrome number?
9. In the figure, all the four shaded semicircles have same area.
    We put a dot in the figure, without looking in to it.
    Find the probability of being dot is
    a. In the shaded semicircles.
    b. Not in the shaded regions



10 . The shaded quadrilateral is formed by joining the midpoints of a rectangle. if we put a dot on the figure without looking into it what is the probability that it falls on the shaded region



11. An equilateral triangle of maximum size is drawn inside the circle.
    a) If the radius of the circle is 'r' then what is the side of the triangle?
    b) If we put a dot inside the figure without looking into it. What is the probability that it will be inside the triangle?
    c) What is the probability that it will be outside the triangle?



12. In a box, there are four slips with numbers 1,2,3 and 4 written on them and another box with two slips with numbers 1 and 2 written on them. If one slip is taken from each box randomly,
    a) What are the possible pairs?
    b) What is the probability that both slips are odd number?
    c) What is the probability that both are even numbers?
    d) What is the probability of getting even in one and odd in the other?
    e) What is the probability of getting same numbers in both slips?
    f)What is the probability of getting the square of the number in one in the other?
13. A box contains 50 balls of black and yellow. Take one ball from this, The probability of getting a yellow ball is $\frac{2}{5}$
    a) How many yellow balls are there in the box ?
    b) What is the probability of getting a black balls ?
14. In a box there are some red beads and blue beads. Total number of beads in the box

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is 24. If a bead is taken from it, without looking, the probability of getting a red bead is $\frac{1}{3}$. Find,

a) The probability of getting a blue bead
b) The number of red beads
c) The number of blue beads

15. In class 10A there are 30 boys and 20 girls and in 10B class there are 25 boys and 25 girls. One student is to be selected from each class.
    a) In how many different ways we can select a pair of students, one from each class.
    b) What is the probability of both being boys?
    c) What is the probability of getting one boy and one girl?
    d) What is the probability of getting atleast one is girl?
16. The total number of students in a class is 40.The probability of selecting a boy from the class is 3/8.Then
    a) What will be the number of boys in the class?
    b) What will be the number of girls in the class?
    c) How many boys should be added to make the probability of boys 1/2?
17. A box containing 50 beads including white and black beads. From this a bead is taken and probability of being it a white one is 2/5. Then
    a) Find the number of white beads ?
    b) Find the number of black beads ?
    c) How many white beads should be added to make the probability of white bead to 50% ?
18.. A box contain five paper slips numbered 1, 2, 3, 4 ,5 and another box contain four paper slips numbered 6, 7, 8 ,9. If one slip is taken from each box .
    a) What is the total number of possible pairs ?
    b) What is the probability of both being odd numbers ?
    c) What is the probability of both being even numbers ?
    d) What is the probability of sum of the numbers being even ?
19. Aperson is asked to say a three digit number .
    what is the probability that the number is a multiple of 111

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5. SECOND DEGREE EQUATION

1. When each side of a square is increased by 2m to make a large square, its area become $144m^2$.
   Side of the new square is .....metre.
   [A. 12, B.10,C. 8, D.14]
2. When each side of a square is reduced by 2m to make a smaller square, its area become $81m^2$.
   Side of the first square is .....metre.
   [A.9, B.11, C.79, D. 83]
3. Which number is added to x^2-6x to get a perfect square?
   [A. 6, B. 36, C. 9, D. -9]
4. The position at which the graph of the polynomial $x^2-10x+25$ meet the horizontal line?
   [A. 5, B. -5, C. 10, D. -10]
5. Perimeter of a non square rectangle is 20 cm. Read the statement below
   i) Its area cannot be exceed 25 cm^2.
   ii) Its sides are always 5 cm.
   iii) One of the side is larger than 5 and the other is smaller than 5.
   iv) One of the side is larger than 10 and the other is smaller than 10.

Now choose the correct answer from those given below.

A. (i) and (ii) are true. B. (i) and (iii) are true.
C. (i) and (iv) are true D. (ii) and (iv) are true.

6. Read the statements below
   Statement 1: The equation $(x+1)^2=49$ has two solutions
   Statement 2: Square 7 and -7 is 49

Now choose the correct answer from those given below.

A. Statement 1 is true, statement 2 is false
B. Statement 1 is false , statement 2 is true
C. Both statements are true and statement 2 is the reason of statement 1.
D. Both statements are true and statement 2 is not the reason of statement 1.

7. Statement 1: One solution of a second degree equation sometimes reject in the practical problem.
   Statement 2: Physical quantity like length, weight, age and so on cannot be negative.

Now choose the correct answer from those given below.

A. Statement 1 is true, statement 2 is false
B. Statement 1 is false , statement 2 is true
C. Both statements are true and statement 2 is the reason of statement 1.
D. Both statements are true and statement 2 is not the reason of statement 1.

8. Statement 1: The maximum area of a rectangle with fixed perimeter occurs when it becomes a square.

Statement 2: Perimeter of a rectangle is 40cm and x is a number then its sides are 10+x and 10-x

Now choose the correct answer from those given below.

A. Statement 1 is true, statement 2 is false
B. Statement 1 is false , statement 2 is true
C. Both statements are true and statement 2 is the reason of statement 1.
D. Both statements are true and statement 2 is not the reason of statement 1.

9. When all the side of a square decreased by 5m, the area became $64m^2$.
   Find length of the first square.
10. When all the side of a square increased by 3cm, the area became 2025 cm^2,
    a)Write the length of second side by taking 'x' as the side of the first square.

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b) Form a second degree equation by using the given data.
c) What is the length of the first square?

11. a) If x is a number then, what is its 60 times?
b) 2016 is obtained when square of a number is added to 60 times of it. Find the number.
12. The sum of the numbers denoted the perimeter and area of a square is 140 .
a) If the length of a side is x, form a second degree equation.
b) Find the length of side.
13. If the product of two consecutive odd numbers is 399, find the numbers.
14. The perimeter of a rectangle is 48cm and its area is 135 cm^2
a) What is the sum its length and breadth?
b) Form a second degree equation?
c) Find the length of sides the rectangle.
15. The product of a number and another number 8 more than it is 105.
a) If the smallest number is 'x' , then find the second number.
b)Which is the least number to be added to make the product a perfect square?
c) Find the numbers.
16. Sum of two numbers is 20 and their product is 95.
i) Write these facts as a second degree equation
ii) Calculate the number.
17. One side of a rectangle is 14cm longer than the other, and its diagonal is 26cm.
a) Write these facts as a second degree equation.
b) Calculate the length of sides of the rectangle.
18. Perimeter of a rectangle is 42cm and its area 500 cm^2
a) If the length is 30+x then what is its breadth?
b) What is the number 'x'?
c) Find the length and breadth of the rectangle?
19. Consider the arithmetic sequence: 10,14,18,...
a) Find the algebraic form of sum of this sequence?
b) What is the sum of first 10 terms ?
c) How many terms of this sequence, starting from the first, are to be added to get 504?
20. One side of a right triangle is 7cm longer than the side perpendicular to it, and its area 60cm^2
Its area is 24 cm^2. Calculate the length of all three sides of the triangle.
21. One side of a right triangle is one centimetre less than twice the length of the side perpendicular to it, the hypotenuses one centimetre more than twice this side.
What are the length of the sides?
22. Sum of two numbers is 6 and their product is 7.
a) One of the number is larger than 3 and the other is smaller than 3. Why?
b) Calculate the numbers.
23. Find two numbers with sum 6 and product 7
24. How many terms of the arithmetic sequence 49,47,45,... starting from the first must we add to get 400 as the sum?
25. The position at which the graph of the polynomial x^2-28x+ 192 meet the horizontal line?
26. In the polynomial $p(x)=x^2+5x-6$, what number should be taken as 'x' to get p(x)=0?

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6.TRIGONOMETRY

1. What is the value of sin x , if x = 60^0

   a. $\frac{1}{2}$, b. $\frac{1}{\sqrt{2}}$ c. $\frac{\sqrt{3}}{2}$ d. $\frac{1}{\sqrt{3}}$

2. Find tan 30^0 using the triangle with angles 30^0, 60^0, 90^0.

   a. $\frac{1}{2}$, b. $\frac{1}{\sqrt{3}}$ c. $\frac{\sqrt{3}}{2}$ d. $\sqrt{3}$

3. Area of a triangle with angles 30^0, 60^0, 90^0 is $8\sqrt{3}$ cm^2, What is the length of its shorter side?

   a. $\sqrt{3}$ cm, b. $\sqrt{8}$cm c. 4 cm d. $4\sqrt{3}$cm

4. A person stands 20 meter away from a tree and sees the top the tree at angle of elevation 45°, what is the height of tree?

   a. $20\sqrt{2}$ m b. $20\sqrt{3}$ m c. 10 m d. 20 m

5. In a day, height of a post and its shadow length are same, What is the angle of elevation of the sun at that instant?

   a. 30^0 b. 45^0 c. 60^0 d. 50^0

6. If x is the moderate angle in a non isosceles right triangle, then what relation between sin x and cos x,

   a. sin x = cos x, b. sin x < cos x

   c. sin x > cos x c. sin x + cos x = 1

7. Diameter of circumcircle of a triangle with a side 4cm and its opposite angle 30^0 is

   a. 4cm b. 2 cm c. $4\sqrt{3}$cm d. 8 cm

8. **Statement 1**: If the perpendicular sides of a right triangle are 3 meter each, then the third side $3\sqrt{2}$ meter

   Statement 2: In any triangle with angles 45^0 , 45^0, 90^0 both the shorter sides have the same length and the length of the longest side is $\sqrt{2}$ times this length.

   a) Statement (1) is true and (2) is false.
   b) Statement (1) is false and (2) is true.
   c) Both the statements are true, (2) is the correct reason of (1).
   d) Both the statement are rue, (2) is not the correct reason of (1).

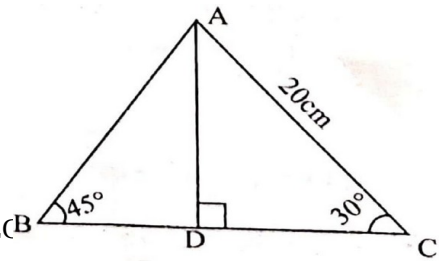
9. **Statement 1**: If the perpendicular sides of a right triangle are 2cm and $2\sqrt{3}$cm, then the third side 4cm

   Statement 2: In any triangle with angles 45^0 , 45^0, 90^0 both the shorter sides have the same length and the length of the longest side is $\sqrt{2}$ times this length.

   a) Statement (1) is true and (2) is false.
   b) Statement (1) is false and (2) is true.
   c) Both the statements are true, (2) is the correct reason of (1).
   d) Both the statement are rue, (2) is not the correct reason of (1).

10. In the figure AC = 20 cm, < B = 45°, and < C = 30^0,The line AD is perpendicular to the side

    a. Find other sides of the triangle ABC



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b. What is the area of the triangle ABC?

[Hint:$\sqrt{3}$ = 1.73]

11.A man standing on the top of a building sees the top of a hill at an elevation of 30^0 and its base at a depression of 45^0. The height of the building is 20 metre

a) Draw a rough figure based on the given details.

b) What is the distance between the hill and the building?

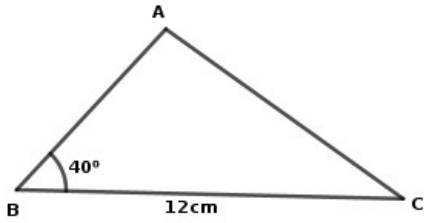
c) What is the height of the hill? [Hint:$\sqrt{3}$ = 1.73]

12. (a)In figure what fraction of AB is the length of perpendicular from A to BC?

(b) If AB =10cm, find the length of perpendicular from A to BC

(c) Find the area of triangle ABC

(d) If two sides of a triangle are 10cm and 12cm and the angle between them is 140°, find the area of triangle



13. In triangle ABC, AB=AC=6cm, <A=120°

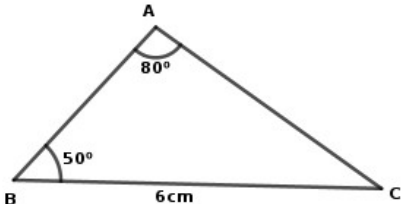
a) Find the length of BC

b) Find the area and perimeter of triangle ABC

14. In the given triangle ABC, <A=50°, <A=80° , BC = 6 cm.

a) Find the circumdiameter of triangle ABC.

b) Find the length of AB and AC



15. A child sees the top of a building far away from him at an elevation of 15°. After walking 40 m towards the building he sees it at an elevation of 30°.

(a) Draw a rough figure indicating the above facts.

(b) Find the height of the building

(c) How much distance should he walk to reach the foot of the building

16. A boy 1.5 metres tall, standing at the edge of a canal sees the top of a tree on the other edge at an angle of elevation of 70° Stepping 20 metres back, he saw it an angle of elevation of 35^0

a) Draw a rough figure based on the given details?

b) What is the height of the tree?

c) What is the width of the canal?

[Sin 35^0= 0.57 , Cos 35^0 = 0.82, tan 35^0 = 0.7 , sin 70^0 =0.94, cos 70^0 = 0.34 , tan 70^0= 2.74]

17. A man standing at the foot of a building, sees the top of a tower 100 metres away at an elevation of 45^0, Climbing to the top of the building it sees at an elevation of 30^0

a) Draw a rough figure based on the given facts.

b) What is the height of the tower?

c) Calculate the height of the building? [Hint:$\sqrt{3}$ = 1.73]

18. If sun is observed in 45^0elevation the length of shadow of a tree is 6m.

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a) find the height of the tree.

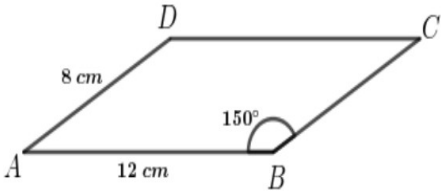
b) When the Sun at elevation is 35^0 ,What is the length of shadow of the same tree? [$\tan 35^0 = 0.7$]

19. A boy of 1.6 metres tall sees the top of a building under construction at an elevation of 30^0. After completing the construction building was 10 metres higher and he sees its top an elevation of 60^0 from the same spot.

a) Draw a rough figure based on the given details ?

b) What is the height of the building ?

20. In parallelogram ABCD ,
AB=12 cm, AD=8 cm ,$<$ B=150^0

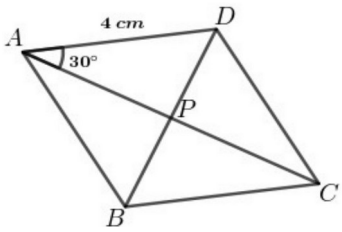


a)What is the measure of $<$ A ?

b)What is the distance from D to the side AB ?

c) What is the area of the parallelogram?

21. The diagonals of a rhombus ABCD intersect at P.
AD=4 cm,$<$PAD=30^0



a) What is the measure of $<$ APD ?

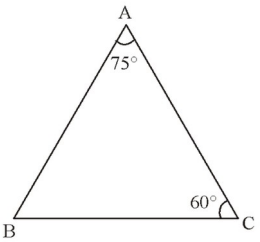
b) What is the length of DP ?

c) What is the length of diagonal AC ?

d) What is the area of the rhombus ?

22. A segment of circle cut of from a circle by a chord of length 12cm and angle in the segment is 130^0. Find the radius of circle.

23. In triangle ABC $\angle$A = 75°, $\angle$C = 60°



a) Find <B.

b) Find AB: BC: AC

c) If AB = $6\sqrt{2}$cm, Find the diameter if its circumcircle

d) find the length of AC, BC

24. The height of a light house from sea level is 25 m. A ship is seen at a depression of 35° at first. After sometimes the ship is seen at a depression of 55°.

(a) Is the ship travelling towards lighthouse? Draw a rough figure?

(b) What is the distance travelled by the ship in this time?

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7. Coordinates

1.Coordinates of of 2 points on a line are (3,7),(3,11)

(i)This line is parallel to x axis

(ii) This line is parallel to y axis

(iii) (7,11) are the coordinates of a point on this line

(iv) (3,15) are the coordinates of a point on this line

Now chose the correct answer from those given below:

(A) (i) and (ii)are true (B) (i) and (iii)are true

(C) (i) and (iv)are true (D) (ii) and (iv)are true

2. The coordinates of 2 points on a line are (2,7), (11,7). Which of the following are the coordinates of a point on this line

A) (2,11) B) (7,2) C) (-2,7) D) (-11,-7)

3. Read the following statements given below

(i) : The y coordinates of points on x axis are zero

(ii): The y coordinates of points on y axis are zero.

(iii): The x coordinates of points on x axis are zero

(iv): The x coordinates of points on y axis are zero.

Now chose the correct answer from those given below:

(A) (i) and (ii)are true (B) (i) and (iii)are true

(C) (i) and (iv)are true (D) (ii) and (iv)are true

4. **Statement A:** The x coordinates of points on a line parallel to y axis are equal

Statement B: Distance between two parallel lines at any where are same

Now chose the correct answer from those given below:

a) Statement A is true and statement B is false.

b) Statement A is false and statement B is true

c) Both statement are true, B is a reason for A

d) Both statement are true but B is not a reason for A

5. **Statement A:** The y coordinates of points on a line parallel to x axis are equal

Statement B: Distance between x axis and points on a line parallel to x axis are equal

Now chose the correct answer from those given below:

a) Statement A is true and statement B is false.

b) Statement A is false and statement B is true

c) Both statement are true, B is a reason for A

d) Both statement are true but B is not a reason for A

6. Find the distance between two points (4,-2) and (4,3)

A) 5 B) 1 C) 6 D) 4

7. Find the distance between x axis and the point (-6,-5)

A) 5 B) 1 C) 6 D) 11

8. Find the distance between y axis and the point (6,-7)

A) 7 B) 1 C) 6 D) 13

9. Find the distance between two points (-8,12) and (-5,12)

A) 3 B) 7 C) 4 D) 13

10. The coordinates of one point on a line which is parallel to y axis are (4,9). Write the coordinates

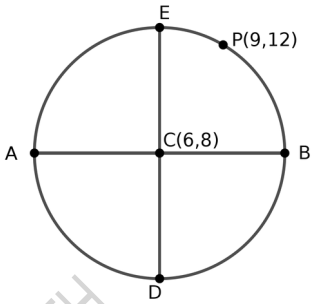
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of two more points on this line

11. The coordinates of opposite vertices of a square are (3,0)and(7,0)
    a) Find the coordinates of other two vertices
    b) Find the length of diagonal
    c) Find the area of square

12. In figure C (6,8) is the centre of the circle and P (9,12) is a point on it.

AB and ED are two diameters of circle which are parallel to the axes



    a) Find the radius of circle
    b) Find the coordinates of A,B, D and E

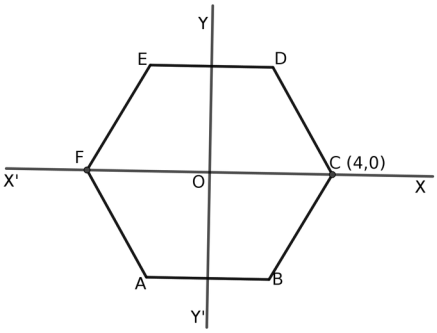
13. The coordinates of a point on a circle with centre at origin 1s (3,4)
    a) Find the radius of circle
    b) Find the coordinates of points at which the circle meets with the axes
    c) find the coordinates of two more points on this circle which are not on the axes

14. The coordinates of opposite vertices of a rectangle whose sides are parallel to the axes are (-3,2), and (9,-3)
    a) Draw a rough figure of the rectangle marking the left right top and bottom corners correctly
    b) Find the coordinates of other vertices
    c) Find the length of sides
    d) Find the perimeter of rectangle

15. The coordinates of centre of a circle is (4,4) and this circle is passing through the origin.
    a) Find the radius of the circle
    b) Check whether the given points are lie on the circle or outside the circle or inside the circle
        i) (7,6) ii) (-2,5) iii) (8,0) iv) (-1,5)

16. Draw x and y axes and mark the points A(0,-1), B(-4,2), C(0,5),D(4,2).Join ABCD .
    a) What is the suitable name for the quadrilateral ABCD?
    b) Find the area of quadrilateral ABCD.

17. In figure O is the origin and ABCD is a regular hexagon. coordinates C is (4,0)



    a) Find the coordinates of F.
    b) Find the coordinates of D E,A and B

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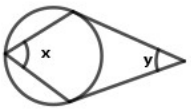
8. TANGENTS

1.In radius of a right triangle is 2 cm and its circumradius is 5 cm then its perimeter is

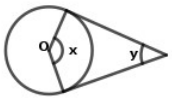
a) 7 cm b)10 cm c) 12 cm d)24 cm

2.Which of the following picture satisfying x+y=180°

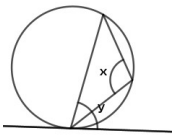
i)



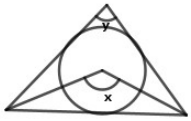
ii)



iii)

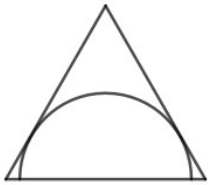


iv)



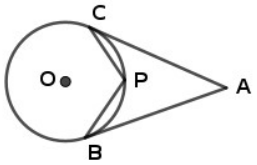
a) i & ii true b) i & iv true c) ii & iii true d) iii & iv true

3.In the figure a semi circle touching two sides of an equilateral triangle of side 4 cm. Then it's radius is



a) $\sqrt{3}$ cm b) $2\sqrt{3}$ cm c) 2 cm d) 4 cm

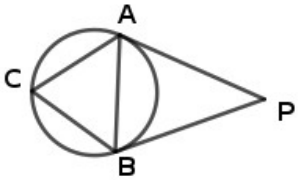
4. In the figure AB, AC are tangents. <A=40° then <BPC= ... ….



a) 70° b) 140° c) 80° d) 110°

5.In the figure PA,PB are tangent to the circle which of the angle given below equal to <PAB?

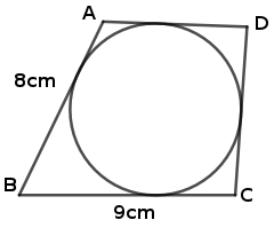
i) <ABC ii) <BAC
iii) <ACB iv) <PBA



a) i & ii true b) i & iv true c) ii & iii true d) iii & iv true

6. In quadrilateral ABCD AB= 8cm BC= 9cm, what are the possible of length of CD and AD?

i) 5 cm ,9 cm ii) 7 cm ,8 cm
iii) 11 cm ,10 cm iv) 8 cm , 7cm



a) i & ii true b) i & iv true c) ii & iii true d) iii & iv true

7. Read the two statements below:

Statement A: The tangent at a point to a circle is perpendicular to the diameter through that point

Statement B: The sum angle at the centre of a circlr between radii and the angle between the tangents at the ends of radii is 180^0.

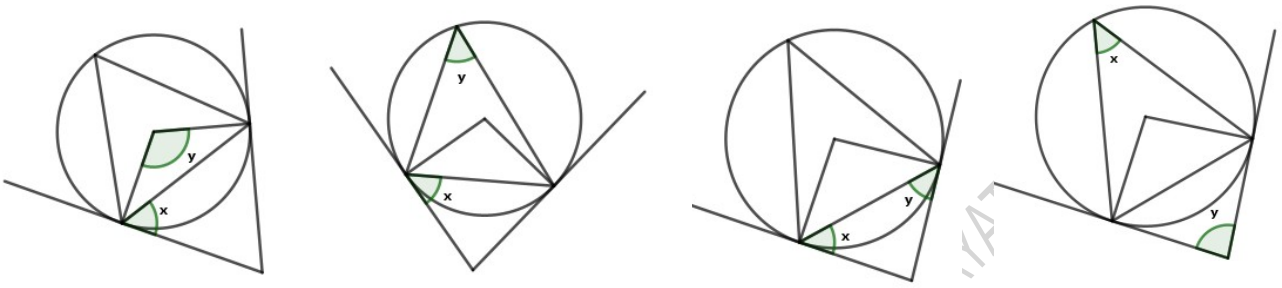
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Now chose the correct answer from those given below:

a) Statement A is true and statement B is false.
b) Statement A is false and statement B is true
c) Both statement are true, B is a reason for A
d) Both statement are true but B is not a reason for A

8. In the pictures below which images are satisfy the equation x = y?

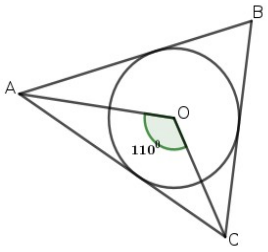
i) ii) iii) iv)



A B C D

a) i & ii true b) i & iv true c) ii & iii true d) iii & iv true

9. In the figure, the two vertices of a triangle are joined to it's incenter.
If < AOB = 110^0 then < ABC = ...



a) 55^0 b) 70^0 c) 40^0 d) 60^0

10 . Read the two statements below:

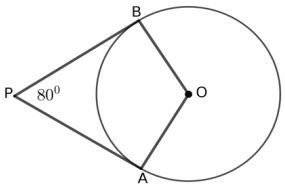
Statement A: The center of the triangle's incircle is on the bisectors of angles of the triangle.

Statement B: Any point at the same perpendicular distance from two sides of an angle lies on the bisector of that angle

Now chose the correct answer from those given below:

a) Statement A is true and statement B is false.
b) Statement A is false and statement B is true
c) Both statement are true, B is a reason for A
d) Both statement are true but B is not a reason for A

11. In the figure O is the centre of circle , PA and PB are two tangents . If < APB = 80^0 then find < AOB

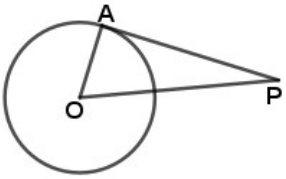


a) 80^0 b) 90^0 c) 100^0 c) 40^0

12. What is the circumradius of an equilateral triangle its inradius radius is 3 cm?
a) 3cm b) 1.5 cm c) 6 cm d) 4.5 cm

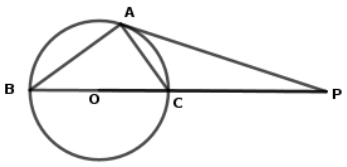
13. What is the area of the triangle with perimeter 50 centimetre and inradius 5 cm?
a) 125cm b) 100cm c) 250cm d) 200cm

14. In the picture O is the centre and PA is the tangent OA=5cm, OP=10cm.
a) What is the length of the tangent PA?
b) What is the measure of <AOP?

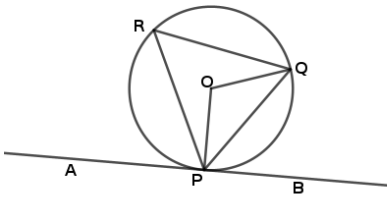


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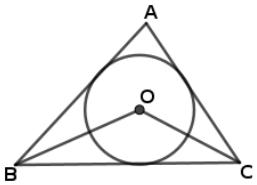
15. In the picture O is the centre and PA is a tangent,If AB=AC then what is <P?



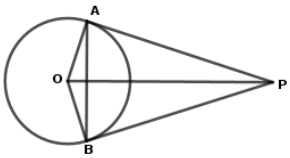
16 .In the figure O is the centre and AB is a tangent,
If <QPB = 50°, <APR = 55° then
a) What is <PRQ and <POQ
b) Find angles of triangle PQR?



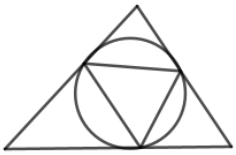
17.In the picture two vertices of a triangle are joined to its centre.
a) If <A=80° then what is <BOC?
b)If <OBC=25° then prove that triangle ABC is an isosceles triangle?



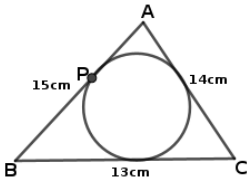
18. In the picture O is the centre PA, PB are tangents, Triangle ABP is an equilateral triangle.
a) if OA= 5 cm then find the length of AB?
b) prove that OP=2r?



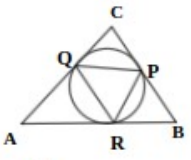
19. In the picture the midpoints of a triangle are joined form a small Triangle
a) prove that the above two Triangles are equilateral triangles?
b) Prove that the sides of small Triangles is half the large Triangle?



20. In the figure sides of triangle ABC touches the circle.
a) What is the perimeter of triangle ABC?
b) find the length of AP?
c) What is the area of triangle ABC?
d) Find the in radius?



21. In the figure sides of triangle ABC touches the circle at the points P, Q, R,<A=40° and<B=60° then
a) Find other two angles of triangle AQR?
b) Find all angles of triangle PQR?



22. If the hypotenuse of a right triangle is 10 cm and its area is 39 cm² then find its in radius?

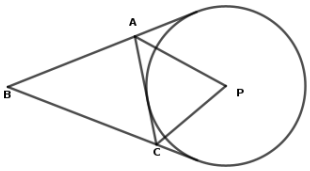
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23.What is the in radius of triangle with sides 13 cm 13 cm 10 cm?

24.In the figure P is the centre of an ex circle of triangle ABC,

a)If <ABC=40°, <BAC=80° then what is the angles of triangle APC?

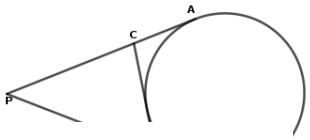
b) If <ABC=x then prove that <BPC=90-x/2 ?



25.In the figure PA, PB, CD are tangents to the circle. If PA=12cm then

a) What is the length of PB?

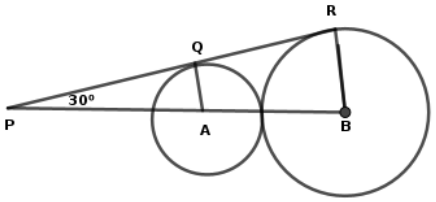
b) What is the perimeter of triangle PCD?



26. In the picture PR touches the circle with the centres A, B and <APQ=30°

a) Prove that radii of circles are in the ratio 1:3?

b) If radius of small circle is 5 cm then what is the length of QR?



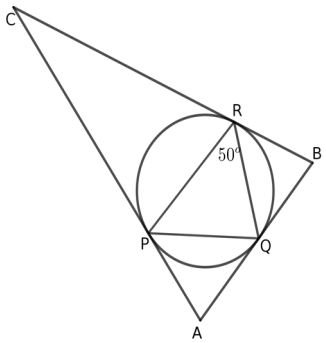
27. Draw a circle of radius 3 cm. draw a triangle with two angles 55°,75° and side touch the circle

28. Draw a circle of radius 2.5 cm mark a point P , 7 cm apart from the centre draw a tangent from P to the circle?

29. Draw a circle of radius 3 cm mark a point P , 7.5 cm apart from the centre draw a tangent from P to the circle?

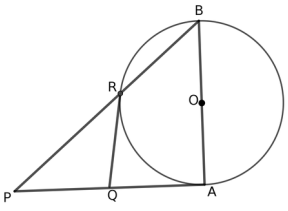
30. Draw a triangle of sides 5 cm ,6 cm and 7 cm and draw it's incircl. Measure the radius.

31. In the picture the sides of the larger triangle are tangents to the circumcircle of the smaller triangle at its vertices. Calculate the angles of triangle APQ, BQR and CPR.



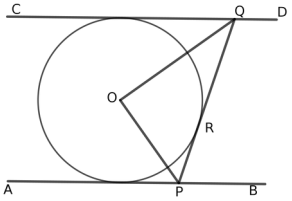
32. In the figure AB is the diameter and PA and QR are the tangents .

Prove that Q is the mid point of PA



33. In the figure AB and CD are two parallel tangents to the circle centred at O. The tangent from the point P on AB meets CD at Q

Prove that POQ is right triangle .



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9. POLYNOMIALS AND EQUATIONS

Choose the correct answer from the given

1. The product of (x+2)(x+8) is .....
   ($x^2 + 6x + 8$; $x^2 + 10x + 16$; $x^2 + 8x + 10$; $x^2 + 12x + 16$)
2. (x +a) (x+b) =.....
   ($x^2 + abx + (a+b)$; $x^2 + (a+b)x + ab$; $x^2 + (a+b) + abx$; $x^2 + (a+b) + ab$)
3. (x- a)(x-b) =.....
   ($x^2 + (b + a)x + ab$; $x^2 + (b - a)x - ab$; $x^2 - (b + a)x + ab$; $x^2 - (b - a)x - ab$)
4. (x + 3)(x + 7) = (x +a) (x+b) ആയാൽ a + b = ...
   (10 , 4 , 21 , 8)
5. $x^2 + 3x + 2$ = ...
   ((x+1)(x+ 1) ; (x+2)(x+ 1) ; (x+3)(x - 1) ; (x+2)(x+ 3))
6. To write the polynomial $x^2 + 5x + 6$ as the product of two first degree polynomials, we find two numbers with
   [(product:5, sum:6), (product:6, sum:5),(product:11, sum:1),(product:6, sum: -5)]
7. The second degree equation representing, a number whose square is 1 more than itself is ....
   ($x^2 + x - 1 = 0$; $x^2 - x + 1 = 0$; $x^2 + 1 = x$; $x^2 = x + 1$)
8. What are the x coordinates of the points where the graph of the polynomial $P(x) = x^2+10x+24$ cuts the x axis?
   (6 , 4 ; 3, 7 ; -4 ,6 ; -4 , -6)
9. What are the coordinates of the point where the graph of the polynomial $P(x) = x^2+6x+9$ cuts the x axis?
   ((0,0) ; (-3, 0) ; (3,0) ; (3,3))
10. If $x^2 -7x+10 = (x-a)(x-b)$ then a+b=.....

[7, -7, 10, -10]

11. Find the product of (x+2)(x+7)
12. Find the product of (x+8)(x-3)
13. Find the product of (x-3)(x-7)
14. Write the second degree polynomial $p(x)=x^2+10x+24$ as the product of two first degree polynomials.
15. Write the second degree polynomial $p(x)=x^2-7x+10$ as the product of two first degree polynomials.
16. Write the second degree polynomial $p(x)=x^2-4x-12$ as the product of two first degree polynomials.
17. $P(x)=x^2+ax+b=(x+3)(x+5)$. Find a and b.
18. Write the second degree polynomial $p(x)=x^2+30x+216$ as the product of two first degree polynomials.
19. If $x^2+7x+12 = (x+a)(x+b)$

a) What is a+b and ab

b) Write $x^2+7x+12$ as the product of two first degree polynomials.

20. Write the second degree polynomial $p(x)=x^2-35x+276$ as the product of two first degree polynomials.
21. Write the second degree polynomial $p(x)=x^2-x-1$as the product of two first degree polynomials.
22. Perimeter of a rectangle is 30cm and its area 54cm^2. Find the sides of the rectangle.

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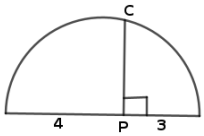
23. Perimeter of a rectangle is 42cm and its area 20cm^2. Find the sides of the rectangle.
24. The product of a number and 5 more than that number is 104. Find the numbers.
25. If $x^2+31x+228 = (x+a)(x+b)$ then,
    a) Find a+b and a-b
    b) Write $x^2+31x+228$ as the product of two first degree polynomials
26. a) $(x + 2)(x+3) = x^2 + 5x + 6$ .What are the two first degree factors of $x^2 + 5x + 6$
    (b) If $(x+a)(x + b) = x^2 + 8x + 15$ find a and b.
c) Write the polynomial $x^2 + 8x + 15$ as the product of two first degree polynomials.
27. The length of a rectangle is 3cm more than its breadth. Its area is 130 cm$^{2.}$ What are the length and breadths?
28. $P(x)=x^2+x-6$ is a second degree polynomial. Write this polynomial as the product of two first degree polynomials.
29. At what values of x, the polynomial $p(x) =x^2-7x-15$ is zero? Write this polynomials as the product of two first degree polynomials.
30. Write the polynomial $p(x)=x^2+4x+1$ as the product of two first degree polynomials and find the solution of the equation $p(x)=0$.
31. Product two consecutive numbers is 462. what are the numbers?
32. Sum of first n terms of an arithmetic sequence is $2n^2+5n$. How many consecutive terms from first should be added to get the sum 1375?
33. At what point does the graph of the polynomial $2x^2-7x-15$ cross the x axis?
34. Find the points at which the graph of the polynomial $3x^2+25x+28$ cross the x axis?
35. How many terms of the arithmetic sequence 5,11,17,... starting from first are to be added to get the sum 320?
36. $P(x)=6x^2+5x-6$. What number should be taken as x to get $p(x)=0$.
37. $P(x)=8x^2-14x-13$. What number should be taken as x to get $p(x)=2$.
38. A rectangle is to be made by bending a 32cm long rod
    a) Can you construct such a rectangle with diagonal 10cm?
    b) How about the rectangle of diagonal 12cm?
    c) And a rectangle of diagonal 16cm?
39. A rectangle garden is to be made with perimeter 42 m and diagonal 15 m.
    a) What is the sum of its length and breadth?
    b) Form a second degree equation and find the dimensions of the rectangle.
40. A rectangle is to be made by bending a 56 m rod.
    a) What is the sum of length and breadth of the rectangle?
    b) If the length of diagonal is 20cm. Find the dimensions of the rectangle.
41. Sum of two numbers is 5 and their product is 4. Find the numbers.

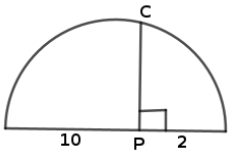
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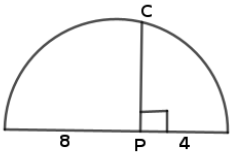
10. CIRCLES AND LINES

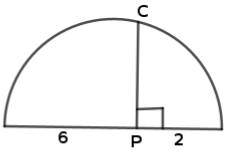
1.AB is the diameter of the circles, in which picture the length of PC is √12?

i) ii) iii) iv)



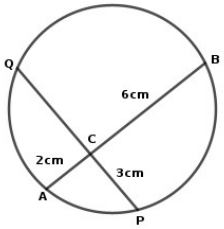






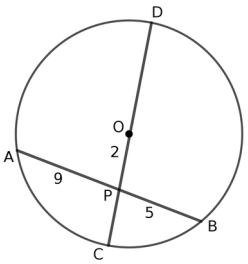
a) i, ii true b) i, iv true c) ii, iii true d) iii, iv true

2. In the figure AC= 2 cm ,
CB = 6 cm and CP = 3 cm
Find the length of CD



a) 4 cm b) 6 cm c) 5 cm d) 3 cm

3. In the figure O centre of circle and CD is the diameter . The chord AB and CD intersect at P
OP = 2 cm ,PA = 9 cm and PB = 5 cm .
Then Find length of diameter .



a) 6 cm b) 7cm c) 9 cm d) $\sqrt{45}$ cm

4. Read the two statements below:

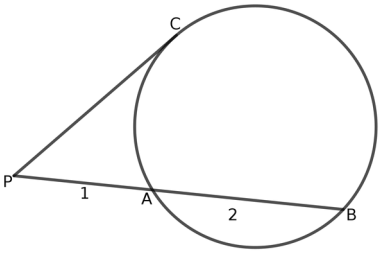
Statement A: When two chords of a circle intersect with in the circle the product of the parts of one chord equal to the product of the parts of the other

Statement B: If two triangle have the same angle then their sides are scaled by the same factor

Now chose the correct answer from those given below:

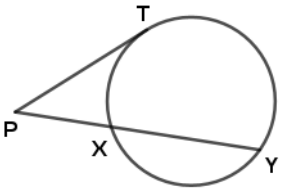
a) Statement A is true and statement B is false.
b) Statement A is false and statement B is true
c) Both statement are true, B is a reason for A
d) Both statement are true but B is not a reason for A

5. In the figure PC is a tangent to the circle from the point P. If P A = 1cm and
AB = 2 cm then what is the length of P C?



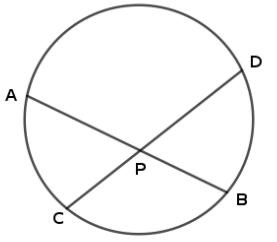
a) 3 cm b) $\sqrt{3}$ cm c) $\sqrt{2}$ cm d) 2 cm

6. In the figure P is a point outside a circle of radius r at a distance d from the centre,d^2-r^2 equal



i) PT ii) PX x PY iii) $(PT)^2$ iv) PX x XY

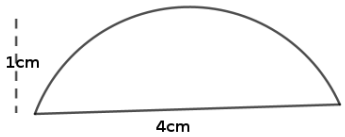
a) i, ii true b) i, iv true c) ii, iii true d) iii, iv true



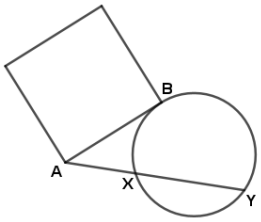
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7. 3.In the figure the chords AB and CD are intersect at P,
a)prove that PA x PB=PC x PD
b) If PA=12 cm, PB=4cm, PD= 8cm then find the length of PC.

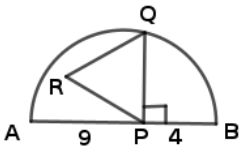
8 . The distance between the ends of a piece of a bangle is 4 cm and its height is 1 cm calculate the radius of full bangle.



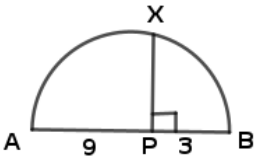
9. In the figure AB is a tangent to the circle AX=4cm, XY=5cm,
a) What is the area of square ABCD?
b) what is the length of the side?



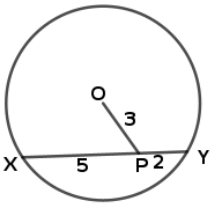
10 . AB is the diameter of the circle AP=9cm, PB= 4cm,
What is the length of side of equilateral triangle PQR? And calculate its area?



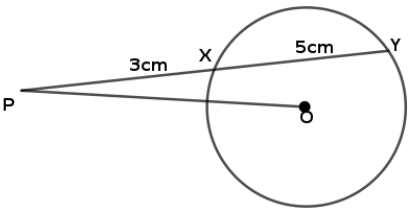
11. AB is the diameter of the circle AP=9cm, PB=3cm,
a) Find the length of PX?
b) Draw a line of length 3√3.
c) Draw a line of length 6√3.



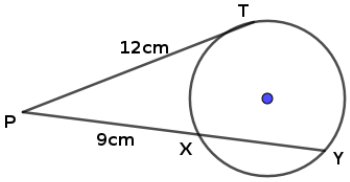
12. In the figure O is the centre of the circle OP=3cm, XP=5cm,
PY=2cm then what is the radius of the circle?



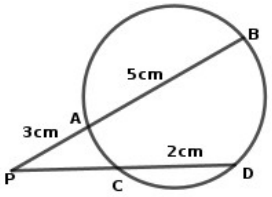
13. In the picture P is the point 7 cm apart from the centre of the circle,PX=3cm, XY=5cm, what is the radius of the circle?



14. PT is the tangent PT =12 cm, PX=9 cm, find the length of X.



15. In the picture PA=3 cm, AB=5 cm, CD=2 cm find the length of PC.

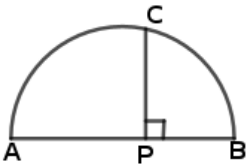


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16. In the picture AB is the diameter

a) If PB =1 cm BC =5 cm then Find the length of AP.

b) If PC= x and PB= 1 cm then AP=.....



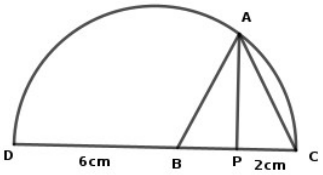
17. Draw a rectangle of sides 5 cm , 4 cm draw a rectangular same area with one side 6 cm.

18. Draw a square of area 12cm²

19.Draw a rectangle of with 6 cm height 4cm. Draw a square of same area.

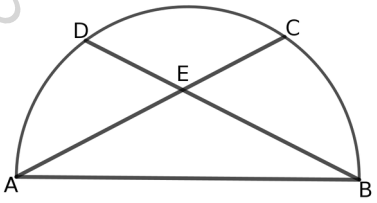
20. Draw an equilateral triangle of sides $\sqrt{8}$ cm

21. In the figure CD is the diameter of semi circle
PD=6cm and PC=2cm . If AB=AC then
find the sides of triangle ABC.



22. Draw a rectangle of area 20 cm^2 and one side 7cm

23. In the figure AB is the diameter of semicircle .
The chords AC and BD intersect at E
prove that AC x AE + BD xBE = AB^2



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11. GEOMETRY AND ALGEBRA

1. The slope of a line is 2.
Which of the following are correct
i) As the x coordinate increases by 1, the y coordinate increases by 2
ii) As the x coordinate increases by 2, the y coordinate increases by 1
iii) As the x coordinate decreases by 1, the y coordinate decreases by 2
iv) As the x coordinate decreases by 2, the y coordinate decreases by 1

(A)(i) and (iii) are true (B)(ii) and (iii) are true

(C)(ii) and (iv) are true (D)(i) and (iv) are true

2. Read the statements below

Statement A: On the line joining (2,5) and (6,7), as we move from any point to another, the change in the y coordinates is half times the change in the x coordinate

Statement B: For any two points on a line , not parallel to either axis, the change in the y coordinates is the change in the x coordinate multiplied by a fixed number

Choose the correct answer from those given below

(A)Statement A is true and statement B is false
(B)Statement B is true and statement A is false
(C) Both the statements are true and statement B is a reason for statement A
(D) Both the statements are true and statement B is not a reason for statement A

3. Read the statements below

Statement A: Slope of the line joining (2,8) and (4,9) is $\frac{-1}{2}$

Statement B: For any two points on a line not parallel to either axis,slope is the the ratio of the change in y coordinate to the change in x coordinate

Choose the correct answer from those given below

(A)Statement A is true and statement B is false
(B)Statement B is true and statement A is false
(C) Both the statements are true and statement B is a reason for statement A
(D) Both the statements are true and statement B is not a reason for statement A

4. The equation of a line is x-y=0

(i) The line is parallel to the axes
(ii) The X coordinate and y coordinate are same for every point on this line.
iii) This line passes through the origin
(iv) The slope of this line is -1

(A)(i) and (ii) are true (B)(ii) and (iii) are true
(C)(iii) and (iv) are true (D)(i) and (iv) are true

5. The equation of a circle is $x^2+y^2=25$. Which of the following are correct

(i) The centre of the circle is origin
(ii) The radius of this circle is 5

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(iii The circle meet x axis only at (0,5)

(iv) The coordinates of a point on this circle is (5,5)

a) i & ii true b) i & iv true

c) ii & iii true d) iii & iv true

6.Equation of the line passing through the point (1,2),(5,6) are

a) x=y+1 b) y=2x

c) y=x+1 d) y=x-1

7.Which among the points is on the line y=2x+1

i) (1,3) ii) (3,8)

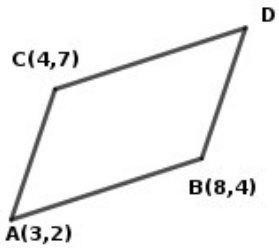
iii) (0,1) iv) (0,-1)

a) i & ii true b) i & iv true

c) ii & iii true d) iii & iv true

8. In a line when the x coordinate increased by 4 ,the y coordinate decreased by 3. Find the slope of this line

a) $\frac{4}{3}$ b) $\frac{-4}{3}$ c) $\frac{3}{4}$ d) $\frac{-3}{4}$



9.In figure ABCD is a parallelogram.

(i)Find the coordinates of D

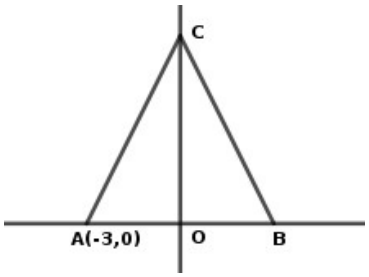
(ii) find the coordinates of point at which the diagonals meet each other

10.A(2,3),B(7,5),C(8,9),D(3,7) are the vertices of the quadrilateral ABCD

a) Find the coordinates of midpoints of the diagonals

b)Prove that ABCD is a parallelogram?

11.In figure ABC is an equilateral triangle and O is the origin.



(i) Find the coordinates of B

(ii) Find the coordinates of C

iii) Find the perimeter of the triangle

(iv) Find the area of the triangle

12. A(3,2) and B(13,7) are the end points of a line AB. Find the coordinates of point P on this line such that AP:PB = 3:2

13. The coordinates of endpoints of the diameter of a circle are (-3,1),(5,7).

(i) Find the coordinates of the centre of circle

(ii) Find the radius of the circle

(iii) Find the equation of this circle

14. The coordinates of 2 points on a line are(2,3) and (5,7).

(i) Find the coordinates of two more points on this line.

(ii) Find the slope of the line.

(iii) Find the equation of the line

15. As we move from one point on the line to another Y- coordinates is increased by to 2 and X- coordinate is increased by 3.

a) Find the slope of the line?

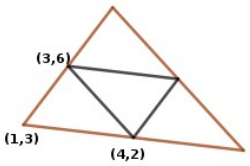
b)If (-1,2) is a point on this line then write other two points on this line?

c)Find the relation between X coordinate and Y co-ordinate of all these points on the line

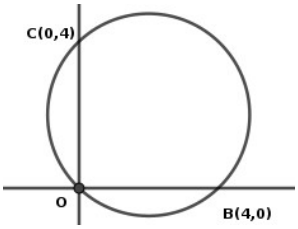
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16. A(4,5),B(9,3) C(6,-1) D(1,1) are the coordinates of the quadrilateral ABCD
    a)Find slope of the diagonals AC and BD?
    b) Find the coordinates of the intersecting points on the diagonals?
17. The coordinates of a point on a line which is passing through the origin are(3,-3)
    (i) Find the slope of this line
    (ii) Prove that the sum of x coordinate and y coordinate of every point on this line is zero.
18. The coordinates of two points on a line are (-2,1) and (1,6)
    (i) Find coordinates of one more point on this line
    (ii) Find the slope of this line
    (iii) Prove that if (x y) is coordinates of point on this line then (x+3,y+5) is also the coordinates of a point on this line

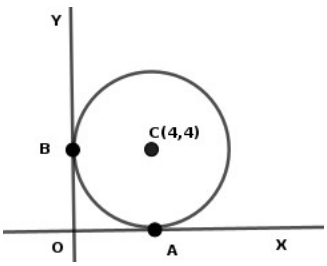
19. The coordinates of 2 points on a line are (1,5),(2,8)
    (i) Find the slope of this line
    (ii) Prove that (x ,3x+2) is also the coordinates of a point in this line
20.The sides of the larger triangle in the picture are parallel to the sides of the smaller triangle.



    a)Calculate the coordinates of the other two vertices of the larger triangle?
    b)Calculate the coordinates of the third vertex of the smaller triangle?
21. In figure O is the origin.



    (i)Find the radius of the circle
    (ii) find equation of the circle
22. The coordinates of centre of a circle is (3,3) and this circle is passing through the origin .
    (i) Find the radius of the circle
    (ii) Find the equation of the circle
    iii) Find the coordinates of the points at which the circle intersect with axes
23. The coordinates of centre of a circle is (0,3) and this circle is interesting y axis at (0,8)
    (i) Find the coordinates of the other end of the diameter through (0,8)
    (ii) Find the radius of the circle.
    (iii) Find the equation of the circle
    (iv) Find the coordinates of the points at which the circle intersect with x axis and y axis
24. The coordinates of centre of a circle is (9,3) and it's radius is 5 unit. Will this circle meet with y axis? Why?
25. The equation of a circle is $x^2+y^2-8x-6y+9=0$.
    (i) Find the coordinates of the centre of the circle
    (ii) Find the radius of the circle
26. In figure the circle with the centre C(4,4) touches x-axis and y-axis at A and B



    (i) Find the coordinates of A and B
    (ii) find the radius of the circle
    (iii) find the equation of the circle
27.If (X_1,Y_1) and (X_2,Y_2) are the coordinates of any two points on a line prove that $(2X_2-X_1, 2Y_2-Y_1)$ are also the coordinates of a point on this line
28. (2,1),(6,7) are coordinates of 2 points on a line.

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(i) What is the relation between change in y coordinates of 2 points on this line and change in their x coordinates

(ii) Check whether the point (14,19) is on this line or not

(iii) the x co-ordinate of one point on this line is 8. Find its y coordinate

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12. SOLIDS

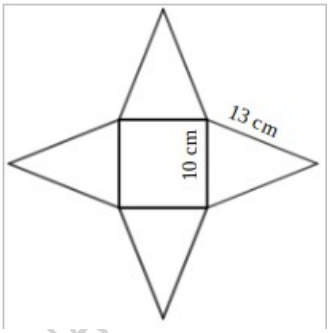
Select the correct answer from those given

1). A square pyramid is made by using a square paper of area 100 cm² and three Isosceles triangles of area 50 cm² each. What is the total surface area of the pyramid?

(150 sqcm; 200 sqcm ; 250 sqcm ; 300 sqcm)

2). The open figure of a square pyramid is given in the picture . What is the sland height of the square pyramid?

(10 cm, 13 cm, 12 cm, 5 cm)



3) all edges of a square pyramid are of equal length. Total length of all edges is 80 cm. What is the slant height of the pyramid?

(5 cm ; 5√3 cm ; 5√2 cm ; 10√3 cm)

4) Base edge of a square pyramid is 12 cm and sland height is 10 cm. What is the height of the pyramid?

(8 cm; 6√2 cm ; 5√2 cm ; 6cm)

5) Base area of a square pyramid is 100 centimetre square and height 15 cm, what is the volume of the pyramid?

(1500 cm^3 ; 4500 cm^3 ; 750cm^3 ; 500 cm^3)

6) From a circle of radius 18 cm. A sector of Central angle 120° is cut and rolled into a cone. What is the slant height of the cone?

(6 cm ; 9 cm; 18 cm ; 36 cm)

7) A circle of radius 15 cm is divided into five equal sectors, one sector is rolled into a cone. What is the radius of the cone?

(15cm ; 10 cm ; 5 cm ; 3 cm)

8. A cone of base radius 10 cm is made by rolling up a sector of a circle of radius 15 cm. What is the central angle of the sector?

(240°, 120°, 72°, 36°)

9. A circle of area 240π cm² is cut into three equal sectors. And rolled into three cones. What is the sum of curved area of cones?

(80 π cm² ; 160π cm² ; 120 π cm² ; 240π cm²)

10. Volume of a wooden cylinder is 300 cm^3, what is the volume of the largest cone which can be carved out of it?

(300 cm^3 ; 150 cm^3 ; 100 cm^3; 75 cm^3)

11) Surface area of a sphere is hundred centimetre square. It is cut into two equal hemisphere. What is the surface area of each hemisphere?

(25 cm² ; 50 cm², 75 cm², 100 cm²)

12) What is the volume of a sphere with radius 3 cm?

(9 π cm^3; 18 π cm^3 ; 27 π cm^3 ; 36 π cm^3)

13) A sector is rolled to make a cone,

i) Radius of the cone is equal to radius of the sector.

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ii) Curved surface area of the cone is equal to area of the sector
iii) Height of the coin is equal to radius of the sector
iv) Base perimeter of the cone is equal to arc length of the sector

a) i , ii true b) i , iv true c) ii, iii true d) ii , iv true

14) Read the two statements below

Statement A: - A square pyramid cannot be made using four isosceles triangles with two sides, 5 cm and one side 8 cm.

Statement B: In any square pyramid slant height is always more than half of the base edge.

Now chose the correct answer from those given below:

a) Statement A is true and statement B is false.
b) Statement A is false and statement B is true
c) Both statement are true, B is a reason for A
d) Both statement are true but B is not a reason for A

15. Read the two statements below

Statement A: - The volume of a square prism is 900 cubic centimeters. The volume of a square pyramid with the same base and height is 300 cubic centimeters.

. **Statement B:** The volume of a square pyramid is one-third that of a square prism of the same base and height.

Now chose the correct answer from those given below:

a) Statement A is true and statement B is false.
b) Statement A is false and statement B is true
c) Both statement are true, B is a reason for A
d) Both statement are true but B is not a reason for A

16. Read the two statements below

Statement A: - If a cone is made by rolling up one-third of a circle , then its radius also one-third of radius a circle

. **Statement B:**In a circle, the circumference and radius are in the same ratio

Now chose the correct answer from those given below:

a) Statement A is true and statement B is false.
b) Statement A is false and statement B is true
c) Both statement are true, B is a reason for A
d) Both statement are true but B is not a reason for A

17. The ratio between the curved surface areas of two cone is 1:3. If the curved surface area of the second cone is 720 square centimeters, what is the curved surface area of the first cone?

a) 360 cm^2 b) 180 cm^2 c) 240 cm^2 d) 300 cm^2

18. If the ratio between the radii of two spheres is 2:3, what is the ratio between their volumes?

a) 8:18 b) 4:6 c) 4:9 d) 8:27

19. A metallic sphere is melted and recast into eight small spheres of radius 8 cm. What is the ratio of radii of larger and smaller spheres?

20. What is the surface area of a square pyramid made by using a Square of sides, 18 cm and four isosceles triangles with two sides, 15 cm

21 What is the ratio between base edge , height and slant height of a square pyramid, if all the edges are equal length.

What is the volume of a square pyramid in this type, if base edge is 8 cm?

22. A square pyramid of maximum size is carved out from A Square Priam with base edges, 18 cm and height 12 cm.

a. What is the volume of the square prism?

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b. What is the volume of the square pyramid?
c. What is the surface area of the square pyramid?

23. A cone is made by using a sector of central angle 288° from circle of radius 25 cm.
    a) what is the slant height of the cone?
    b) Calculate the curved surface area of the cone?
    c) what is the volume of the cone?
24. Base radius of a cone is 12 cm and height 16 cm
    a. What is slant height of the cone?
    b. What is the radius and central angle of the sector used to make this cone ?
    c. What is the current surface area of the cone?
25. Volume of a wooden cylinder is 300 cm^3 . What is the volume of the largest cone carved out of it?
26. surface area of a solid sphere is 400π cm^2,
    a. What is the radius of the sphere?
    b. it is melted and cast into small sphere of radius 1 cm, how many spheres can be made?
    c. A solid metal hemisphere of surface area 300 pie cm Square is melted and recast into cones of radius 1 cm. How many spheres can be made?
27. A solid metallic hemisphere of radius 12 cm is cut into two equal hemispheres,
    a) What is volume of each hemisphere?
    b) How many spheres of radius 3 cm can be made by melting and casting it
    c) The second hemisphere is melted and cast into 32 cones of base radius 3 cm. Then what is the height of the cone?
28 . The slant height of a square pyramid is13 centimes and its surface area 360 $cm^{3.}$ What is its volume ?
28. A circle of radius 12 cm is cut into two sector with central angles in the ratio2:3 . And rolled into two cones.
    a) what is the ratio of their radii?
    b) what is the ratio of bass perimeters?
    c) what is the ratio of curved surface areas?
    d) what is the sum of radii?
    e) what is the sum of base perimeters?
    f) what is the sum of curved surface areas?
29. Prove that lateral surface area of largest square pyramid, which is curved curved out of a cube is root five times its base area
30. a) If base edge and slant height of a square pyramid are equal, what is the relation between base area and lateral surface area?
    b) If base edge and slant height of a square pyramid are 10 cm. What is the surface area of the pyramid?
31. If surface areas of two hemispheres are in the ratio is 4:9. If the volume of the smaller hemisphere is 800 cm cube, what is the volume of the bigger?
32. 27 small sphere of radius 2 cm are melted and recast into a big sphere. What is the radius of the big sphere
33 What is the ratio between the volume of cone of same base radius and height , hemisphere and sphere of same radius that of the cone.
    b) A metal sphere of radius 8 cm is melted and recast into cones of base radius and height 8 cm. How many cones can be made?
35. A wooden toy in the shape of cone with the base radius 30 centimetre and height 40 centimetre. What is the total cost of painting 100 such toys ,at 50 rupees per square meter ?

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13. STATISTICS

1: What is the median of the numbers given below?

1, 5 , 4 ,3 , 7 , 6 , 9

(A) 35 (B) 5 (c) 3 (D) 10

2: What is the mean of the numbers given below?

2, 4, 1, 8, 9

(A) 1 (B) 4 (c) 24 (D) 9

3:The mark scored by 10 students in an exam given below. Find the median of the marks

11,32,33,35,39,41,45,47,48,49

(A) 35 (B) 36 (C) 38 (D) 40

4:The weight of seven children is given below. Find the mean of the weight

35,43,38,45,32,44,42

5:Statement 1 The median of 5,10,15,20,25 is 15
Statement 2 Median and mean of arithmetic progression is equal

A :Statement one is true and statement 2 is wrong
B :Statement 1 is wrong and statement 2 is true
C :Statement one and two are true and one is the reason of two.
D :Statement 1 and 2 are true but statement 2 is not the reason of statement 1

6: Daily income of the workers in a factory is given below.

Daily income	No of workers
225	4
250	7
270	9
300	5
350	3
400	2

Find the mean of the daily income

7:The mark scored by the students in a class given below.

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20 ,22, 18 ,17 ,21, 23
Find the mean

8: A table of the students in maths club is given below.

Height	No of students
120-130	2
130-140	7
140-150	10
150-160	5
160-170	1
total	25

a) When arranged in ascending order of height, the height of which numbered child is considered the median height?"

b) Find the mean of the height

9:The table of the wage of 99 workers in a factory given bellow

Daily wage	No of students
500-600	8
600-700	13
700-800	20
800-900	25
900-1000	19
1000-1100	14

a) If the workers are arranged on the basis of their daily income, the income of which numbered worker is taken as the median income?"

b) Which is the median class

c) what is the median income

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10 : Some numbers are written below in ascending order. If the median is 15.5, what is the value of x ?

11: A table classifying the students of a class based on their height is given below. What is the median height?

Height (cm)	No of students
140-145	5
145-150	8
150-155	12
155-160	16
160-165	11
165-170	5
170-175	3

12:The marks obtained by the students of a class are given below.

mark	No of students
30	2
41	4
32	3
47	5
24	8
31	3

a) The marks of which student is the median?

b) Calculate the median of the marks."

13: The monthly income of 31 families in a panchayat is given below

Monthly income	No of family
10000-11000	7
11000-12000	4
12000-13000	6
13000-14000	8
14000-15000	4
15000-16000	2

a) The income of which numbered family is the median?

b) Calculate the median of the monthly income ?

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LEAP-26

ENGLISH MEDIUM



SSLC QUESTION POOL